



Khulna University, Khulna
Mathematics Discipline

Year: 4th, Term-II Examination; Session: 2017-2018(Special)
Course No: Math-4203; Title of the Course: **Mathematical Modeling**
Full Marks: 60, Time: 03 Hours

28/22/22
218024

- The figures in the right margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | Mar |
|--|-----|
| 1. (a) Explain modeling and mathematical modeling. Discuss the main stages of mathematical modeling with diagram. Discuss some applications of mathematical modeling in different rising areas. | 05 |
| (b) Define discrete and continuous models, differential and difference equation models. Derive a mathematical formulation for the conservation law in modeling. | 05 |
| 2. (a) Discuss the logistic population model with allee effect. | 06 |
| (b) A tank initially contains 100gal of pure water. Starting at $t = 0$, a brine containing 4lb of salt per gallon flows into the tank at the rate of 5gal/min. The mixture is kept uniform by stirring and the well-stirred mixture flows out at the slower rate of 3gal/min. How much salt is in the tank at the end of 20 min? | 04 |
| 3. (a) Discuss the Richardson Arms Race model. | 06 |
| (b) The fish growth model by Von Bertalanffy is given by $\frac{dF}{dt} = \alpha F^{3/2}(t) - \beta F(t)$, where $F(t)$ denotes the weight of fish and α, β are positive constants. Discuss the stability at their equilibrium points. | 04 |
| 4. (a) Define Lipschitz continuity. State and prove the uniqueness theorem of an initial value problem. | 05 |
| (b) Discuss the method of successive approximation. Apply this method to solve the initial value problem $\frac{dy}{dx} = e^x + y^2$, $y(0) = 0$. | 05 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

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|--|----|
| 5. Discuss the basic assumptions of infectious diseases. Define basic reproduction number. Formulate an epidemic model with horizontal transmission and test the stability at disease free equilibrium points. | 10 |
| 6. Define prey, predator and predation. Discuss a two-species prey-predator model with mathematical formulation. Test the stability at equilibrium points. Also find the trajectories of the model. | 10 |
| 7. (a) Formulate a model with delay differential equation. | 05 |
| (b) Solve the DDE $\frac{dx(t)}{dt} = -x(t-1)$, $t > 0$, with historical function: $x(t) = 1$, $-1 \leq t \leq 0$. | 05 |
| 8. (a) State and solve the Mckendrick-Von Foerster partial differential equation. | 06 |
| (b) Find u_n if $u_0 = 0$, $u_1 = 1$ and $u_{n+2} + 16u_n = 0$. | 04 |

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Khulna University, Khulna

Mathematics Discipline

Year: 4th, Term-II Examination; Session: 2018-2019

Course No: Math-4203; Title of the Course: Mathematical Modeling

Full Marks: 60, Time: 03 Hours

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- Use separate answer script for each section.

Section-AThere are **FOUR** questions in this section. Answer any **THREE** questions

Mark:

- Discuss the latest developments in mathematical modeling. 04
 - State the Pearl-Verhulst population model and explain per capita growth rate, population growth rate and population trajectory and analyze the stability of the model at equilibrium points. 06
- What is the basic assumption of Malthusian growth model? Derive an exponential growth model and discuss its solution for different growth parameters along with the limitations of such model. 05
 - Define threshold parameter and carrying capacity. Derive a bounded growth model with carrying capacity. Solve the model and comment on its plausibility. 05
- Sketch the bifurcation diagram and characterize the bifurcations for the differential equation, $\frac{dx}{dt} = \mu x - x^3$ in which x is the state variable and μ is the bifurcation parameter. 06
 - Discuss the SIR epidemic model. 04
- Describe an artificial kidney machine. Construct a mathematical model for kidney dialysis. Validate the model and discuss its solution. 05
 - Identify the model $\frac{dx}{dt} = x[1-x-\alpha y]$, $\frac{dy}{dt} = \rho y[1-y-\beta x]$ and analyze the stability at equilibrium points. 05

Section-BThere are **FOUR** questions in this section. Answer any **THREE** questions

- Establish a discrete mathematical model based on Newton's law of cooling and solve it. 06
 - Consider the price model $p_{n+1} = \frac{1}{p_n} + \frac{p_n}{2} - 1$. Find the two equilibrium points and determine their stability. 04
- Describe the background of Lotka-Volterra two-species model. Formulate such model and discuss its solution in absence of each other. Also find the trajectories of that model. 10
- Formulate a mathematical model based on supply-demand with delay differential equation. 05
 - Consider the linear system $x' = -x$, $y' = -2y$. Show that the critical point $(0, 0)$ of the system is asymptotically stable. 05
- Solve $U_n = K + U_{n-1}$. 04
 - Discuss the stability analysis of non-linear difference equations. 06

***** THE END *****

Khulna University, Khulna
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Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | Marks |
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| 1. (a) Distinguish between modeling and mathematical modeling and discuss the main stages in modeling process. | 03 |
| (b) Describe some basic principles of mathematical modeling. | 03 |
| (c) Construct a suitable single-species model considering the environmental limitations into account and discuss the solution of the model. | 04 |
| 2. (a) Formulate a carbon dating model and discuss its solution and stability analysis. | 05 |
| (b) Discuss competitive predator-prey models for interacting populations. | 05 |
| 3. (a) Define coupled differential equations. Construct a coupled model of fighting armies between two nations. Discuss the qualitative analysis of the model. | 05 |
| (b) A mathematical model for predicting consumer behavior to decision process of an individual consumer towards a particular product, (say brand X) of a company can be formulated as $\frac{dB}{dt} = b(\mu - \beta B)$, $\frac{d\mu}{dt} = a(B - \alpha\mu) + \gamma c$, where $B(t)$ is the buying level for brand X , $\mu(t)$ is motivation towards brand X , $C(t)$ is the level of communication of brand X and a, b, α, β and γ are constants. Solve the model with qualitative analysis. | 05 |
| 4. (a) Sketch the bifurcation diagram for the model $\frac{dx}{dt} = \mu x - x^3$ in which x is the state variable and μ is the bifurcating parameter. | 05 |
| (b) Discuss the logistic population model with allee effect. | 05 |

Section-B

There are **FOUR** questions in these sections. Answer any **THREE** questions

- | | |
|---|----|
| 5. (a) Protein synthesis can be modeled by the equation $\frac{dy}{dt} = \frac{c}{a + lz} - ky$, $\frac{dz}{dt} = ey - fz$, where a, c, e, f, k and l are positive constants, and $y(t)$ is the RNA concentration and $z(t)$ is the enzyme concentration. Test the stability at the equilibrium points. | 05 |
| (b) Consider the price model $P_{n+1} = \frac{1}{P_n} + \frac{P_n}{2} - 1$. Find the two equilibrium points and determine their stability. | 05 |
| 6. Discuss different compartmental models for epidemic diseases with diagrams. Explain the role of basic reproduction number in the spread of epidemic diseases. Formulate a mathematical model when the disease has a negligibly short incubation period and the recovered individual gets permanent immunity. Discuss the solution with stability analysis. | 10 |
| 7. (a) Distinguish between differential equations and difference equations. Formulate a single species logistic model in terms of difference equations and analyze the solution of that model. | 05 |
| (b) A perfect balanced diet model ensuring certain minimum intake of vitamins A, B, and C from two different foods P and Q for a family can be formulated as,
Min $z = 3x_1 + 2x_2$; subject to $7x_1 + 2x_2 \geq 30$, $5x_1 + 4x_2 \geq 20$, $2x_1 + 8x_2 \geq 16$, $x_1, x_2 \geq 0$ where x_1 is the number of units of food P and x_2 , the number of units food Q. Apply linear programming technique to obtain maximum stuff from each food with minimum cost. | 05 |
| 8. Explain the importance of delay in mathematical model of epidemic diseases. Introduce a constant delay in SIRS continuous time model to convert it to a delay SIRS model. Test the stability at equilibrium points and discuss the effect of delay in the model. | 10 |

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- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

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| 1. (a) Distinguish between modeling and mathematical modeling and discuss the main stages in modeling process. | 03 |
| (b) Describe some basic principles of mathematical modeling. | 03 |
| (c) Construct a suitable single-species model considering the environmental limitations into account and discuss the solution of the model. | 04 |
| 2. (a) Formulate a carbon dating model and discuss its solution and stability analysis. | 05 |
| (b) Discuss competitive predator-prey models for interacting populations. | 05 |
| 3. (a) Define coupled differential equations. Construct a coupled model of fighting armies between two nations. Discuss the qualitative analysis of the model. | 05 |
| (b) A mathematical model for predicting consumer behavior to decision process of an individual consumer towards a particular product, (say brand X) of a company can be formulated as $\frac{dB}{dt} = b(\mu - \beta B)$, $\frac{d\mu}{dt} = a(B - \alpha\mu) + \gamma C$, where $B(t)$ is the buying level for brand X , $\mu(t)$ is motivation towards brand X , $C(t)$ is the level of communication of brand X and a, b, α, β and γ are constants. Solve the model with qualitative analysis. | 05 |
| 4. (a) Sketch the bifurcation diagram for the model $\frac{dx}{dt} = \mu x - x^3$ in which x is the state variable and μ is the bifurcating parameter. | 05 |
| (b) Discuss the logistic population model with allee effect. | 05 |

Section-B

There are **FOUR** questions in these sections. Answer any **THREE** questions

- | | |
|--|----|
| 5. (a) Protein synthesis can be modeled by the equation $\frac{dy}{dt} = \frac{c}{a + lz} - ky$, $\frac{dz}{dt} = ey - fz$, where a, c, e, f, k and l are positive constants, and $y(t)$ is the RNA concentration and $z(t)$ is the enzyme concentration. Test the stability at the equilibrium points. | 05 |
| (b) Consider the price model $P_{n+1} = \frac{1}{P_n} + \frac{P_n}{2} - 1$. Find the two equilibrium points and determine their stability. | 05 |
| 6. Discuss different compartmental models for epidemic diseases with diagrams. Explain the role of basic reproduction number in the spread of epidemic diseases. Formulate a mathematical model when the disease has a negligibly short incubation period and the recovered individual gets permanent immunity. Discuss the solution with stability analysis. | 10 |
| 7. (a) Distinguish between differential equations and difference equations. Formulate a single species logistic model in terms of difference equation and analyze the solution of that model. | 05 |
| (b) A perfect balanced diet model ensuring certain minimum intake of vitamins A, B, and C from two different foods P and Q for a family can be formulated as, Min $z = 3x_1 + 2x_2$; subject to $7x_1 + 2x_2 \geq 30$, $5x_1 + 4x_2 \geq 20$, $2x_1 + 8x_2 \geq 16$, $x_1, x_2 \geq 0$ where x_1 is the number of units of food P and x_2 , the number of units food Q. Apply linear programming technique to obtain maximum stuff from each food with minimum cost. | 05 |
| 8. Explain the importance of delay in mathematical model of epidemic diseases. Introduce a constant delay in SIRS continuous time model to convert it to a delay SIRS model. Test the stability at equilibrium points and discuss the effect of delay in the model. | 10 |



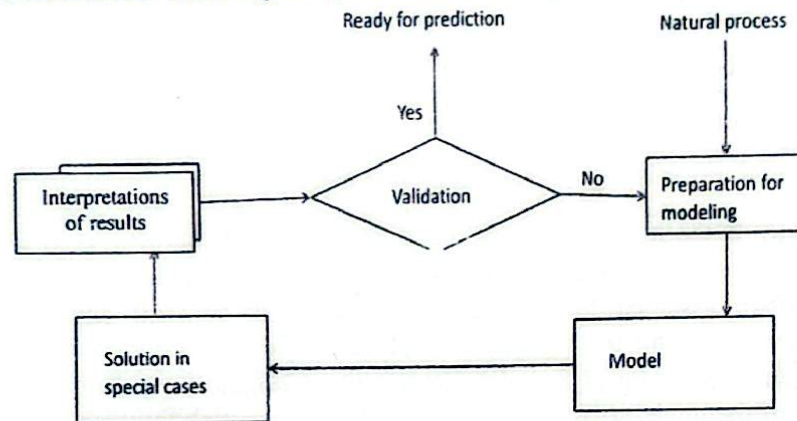
- Figures in the margin indicate full marks. Questions are of equal value.
- Use separate sheet for each section.

Section-A

There are FOUR questions in this section. Answer any THREE questions

Marks

1. (a) Distinguish between modeling and mathematical modeling. Discuss briefly the process of mathematical modeling using the following diagram 03



- (b) 'Conservation law plays a pivotal role in constructing a mathematical model'-Explain with your arguments. 03
- (c) State Newton's cooling/heating law. Suppose that a chicken is removed from an oven, its temperature is then measured at 300°F. Three minutes later, if its temperature is 200°F, then how long will it take for the chicken to cool off to a room temperature of 70°F. 04
2. (a) State the Malthusian growth law for a single species population. Construct an exponential growth model for a single species population. Discuss its solution and limitations, if any. 04
- (b) Explain migration and harvesting in ecology. Suppose migration is considered and let e be the migration rate. Then for harvesting a renewable resource, an ecological model can be formulated as $\frac{dx}{dt} = rx\left(1 - \frac{x}{K}\right) - ex$. Find an analytical solution of the model and discuss the stability at equilibrium points. 06
3. Consider a population of fish that is growing logistically and that is being harvested, assuming the catch of fish per unit effort is proportional to the stock level N . Discuss the model applying high-mortality harvesting and severe overfishing. Also, explain maximum sustainable yield at critical depensation. 10
4. (a) Define linear and non-linear autonomous and non-autonomous dynamical systems. Explain the coupled and non-coupled differential equations. Discuss the equilibrium points and stability at equilibrium points. 04
- (b) A two species competition model with linear autonomous systems can be formulated as $\frac{dx}{dt} = 3x - y$, $\frac{dy}{dt} = -2x + 2y$, where $x(0) = 90$ and $y(0) = 150$ and t is measured in years. (i) Determine the populations $(x(t), y(t))$ at all future times t . (ii) What will be the populations after six months? (iii) When will the species y be eliminated? 06

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

5. (a) Discuss the importance of delay in epidemic model. Incorporating delay in the systems, formulate a basic epidemic SIR model and solve the model with stability analysis. 04
 (b) A nonlinear prey-predator model with Holing type II response function takes the form 06

$$\dot{x}(t) = rx \left(1 - \frac{x}{K} \right) - \frac{sxy}{1 + sHx}, \quad \dot{y}(t) = \frac{esxy}{1 + sHx} - my.$$
 Let s be the search rate, P be the predation rate per predator, f be the fraction of time spent for searching and H be the time needed to handle one prey, then show that $P = \frac{sx}{1 + sHx}$. Also test the stability of the model at the equilibrium points.

6. (a) Solve the following delay differential equation on the interval $[0, 2]$ $\frac{dx}{dt} = x(t-1) + x(t)$, 04
 with initial condition $x(t) = 0$ for all $t \in [-1, 0]$.
 (b) Cooke proposed a delayed vector disease model given by $\frac{dy}{dt} = by(t-\tau)[1 - y(t)] - ay(t)$, 06
 where $y(t)$ is the infected population, b is the contact rate and a is the cure. The discrete time delay τ gives the incubation period before the disease agent can infect a host. Cooke assumed that the total population is constant and scaled so that $x(t) + y(t) = 1$, $x(t)$ being the uninfected population. He also assumed that and infected population is not subject to death, immunity or isolation. (i) Find the steady state solution of the model. (ii) Find the condition for linear stability about the steady state solution. (iii) Obtain the numerical solution for (a) $a = 5.8, b = 4.8(a > b), \tau = 5$ and (b) $a = 38, b = 4.8(a < b), \tau = 5$. (iv) Compare and comment on the graphs.

7. (a) A rod of length 10 cm, whose sides are insulated, is kept at temperature 0°C and 100°C at its ends A and B respectively, until the steady state condition prevails. The temperature at end A is suddenly increases at 20°C and at end B, it is decreased 60°C . formulate a mathematical model of the given situation and obtain the temperature function at time t . 05
 (b) Derive the one-dimensional wave equation considering a homogeneous string of length L , tied at both ends such that the string offers no resistance due to bending. 05

8. Write down some basic features of zoonotic diseases. Is Nipah Virus (NiV) infection a zoonotic disease? Formulate an SEIR model of NiV infection using bilinear incidence rate in absence of permanent immunity and discuss the stability analysis in vein of basic reproduction number R_0 . 10

***** THE END *****

Khulna University, Khulna

Mathematics Discipline

4th Year, Term-I Examination; Session: 2021-2022

Course No: Math-4103

Course Title: Mathematical Modeling

Full Marks: 60

Time: 03 hours

1. The figures in the margin indicate full marks. The questions are of equal value.
2. Use separate sheet for each section.

Section A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

1. (a) Define dynamical systems, modeling and mathematical modeling. Discuss the different processes of mathematical modeling with diagrams. (03)
(b) What is balance principle? Discuss the balance principle in modeling the dynamical systems. (04)
(c) Mention some rising fields and discuss the importance of mathematical modeling to describe the dynamical systems in those fields. (03)
2. Consider a population of fish that is growing logistically and that is being harvested assuming the catch of fish per unit effort is proportional to the stock level N . Discuss the model applying high-mortality harvesting and severe overfishing. Also, explain maximum sustainable yield at critical depensation. (10)
3. (a) What are the basic assumptions of a two-species prey-predator interacting model? Find the exact solution of two-species Lotka-Volterra prey-predator model. (06)
(b) Discuss the nature of the trajectories of the above prey-predator interacting model. (04)
4. (a) The population of a community is known to increase at a rate proportional to the number of people present at a time t . If the population has doubled in 6 years, how long it will take to triple? (04)
(b) What do you mean by radioactive decay and half life of a radioactive substance? (06)
A radioactive isotope has an initial mass 200mg, which two years later is 50mg. Find the expression for the amount of the isotope remaining at any time. What is its half-life?

Section B

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

5. (a) Presently you weigh 169 pounds. You consume x pounds worth of calories each week. Assume your body burns off the equivalent of 3% of its weight each week through normal metabolism. In addition, you burn off $\frac{1}{4}$ pound of weight through daily exercise each week. Find x to one decimal place if you want to weigh between 144 and 146 pounds in 1 year (52 weeks). (04)

- (b) Suppose that each day, 3% of material A decays into material B and 9% of material B decays into lead. Suppose that initially there are 50 grams of A and 7 grams of B . (06)

(i) Formulate a discrete dynamical system to model this situation. How much of each material will be left after 5 days?

(ii) Make a graph of $A(n)$ and $B(n)$ for n going from 0 to 50, and observe how they behave.

(iii) Suppose that after 30 days there are 20 grams of material B left, but there were only 10 grams of B to start with. How many grams of material A were there to begin with, to the nearest gram?

6. (a) Define linear and nonlinear autonomous and non-autonomous dynamical systems. What is equilibrium points? (04)

- (b) Find the equilibrium point and nature of the critical point, also determine the stability of the point and sketch the trajectory near the equation of the system: (06)

$$\begin{aligned}\frac{dx}{dt} &= 2x + 7y \\ \frac{dy}{dt} &= 3x - 8y\end{aligned}$$

7. (a) A rod of length 10cm, whose sides are insulated, is kept at temperature 0°C and 100°C at its ends A and B respectively, until the steady state condition prevails. The temperature at end A is suddenly increased to 20°C and at end B , it is decreased to 60°C . Formulate a mathematical model of the given situation and obtain the temperature function at any time t . (05)

- (b) Derive the one-dimensional wave equation considering a homogeneous string of length L , tied at both ends such that the string offers no resistance due to bending. (05)

8. (a) Michael is taking an exam in order to become a volunteer firefighter. The exam has 10 essay questions and 50 short questions. He has 90 minutes to take the exam. The essay questions are worth 20 points each and the short questions are worth 5 points each. An essay question takes 10 minutes to answer and a short question takes 2 minutes. Michael must do at least 3 essay questions and at least 10 short questions. Michael knows the material well enough to get full points on all questions. He attempts and wants to maximize the number of points. He will do how many short and essay questions? (06)

- (b) Formulate an integro-differential equation to express the population with logistic growth and toxicity of population. (04)

**** End ****

Khulna University, Khulna
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98

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

37

Marks

- (a) Define dynamical systems and mathematical modeling of dynamical systems. Discuss the different processes of constructing a mathematical model with schematic diagram. **03**

(b) Explain the conservation law in dynamical modeling. How does a mathematical model obey the conservation law? **03**

(c) State and formulate the cooling law of an object. A dead body is found at 6.30 AM with temperature 18°. At 7.30 AM, the body temperature is 16°. Suppose the surrounding temperature is 16° and the alive people's temperature is about 37°. Estimate the dead time. **04**
- Find the equilibrium point and determine the stability of the points and draw the trajectories near the equilibrium of the system $\frac{dx}{dt} = x\left(1 - \frac{1}{2}y\right)$, $\frac{dy}{dt} = y\left(\frac{1}{4}x - \frac{3}{4}\right)$. **10**
- (a) Define Allee effect and crowding effect of a biological model. Solve the model $\frac{dx}{dt} = rx\left(1 - \frac{x}{k_1}\right)\left(\frac{x}{k_2} - 1\right)$ analytically where $r = 0.005$, $k_1 = 100$ and $k_2 = 20$. Discuss the stability with allee effect, if any. **05**

(b) A model for species cooperation is given by $x' = -2x + 4y$, $y' = x - 2y$. Discuss the nature of cooperation and determine the predicted future population. What happens as $t \rightarrow \infty$? **05**
- (a) Discuss a model of wildlife ecology. Construct a two-species non-linear model who in the absence of each other increase in a Malthusian manner but who kill each other when competing for food. Determine the equilibrium points and solve for the general trajectories. **05**

(b) Discuss some basic assumptions of constructing endemic model. Formulate an endemic model with nonlinear incidence rate considering loss of immunity. Test the positivity and stability at disease free equilibrium points in terms of basic reproduction ratio. **05**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

5. What do you mean by difference equation model? Explain the stability near the equilibrium of the model $\frac{dx}{dt} = r(1-y)x$, $\frac{dy}{dt} = m(x-1)y$. Also show the effect of the parameters r and m . 10
6. Discuss the importance of delay in biological model. Formulate a SIR delay model for non-communicable diseases. Discuss the solution and stability of that model with basic reproduction number R_0 . 10
7. Consider the prey-predator model with functional response $\frac{dx}{dt} = rx\left(1 - \frac{x}{k}\right) - bxy$, $\frac{dy}{dt} = cxy - dy$ where x is prey and y is predator. Examine the positivity and analyze its stability. 10
8. (a) Define convex function and give an example to show that a convex function has a global minima. Solve the continuous optimization model 04
Maximize $V(x, y, z) = xyz$
Subject to
 $2x + 2y + z \leq 84$; $x > 0$, $y > 0$, $z > 0$.
- (b) A profit maximization model can be formulated as, 06
Maximize $Z = 30x_1 + 30x_2$
Subject to
 $6x_1 + 3x_2 \leq 1200$, $6x_1 + 9x_2 \leq 1800$; $x_1, x_2 \geq 0$
where x_1 is the number of desks and x_2 the number of tables produced in a factory.
Find the optimal solution of the model by applying linear programming technique.

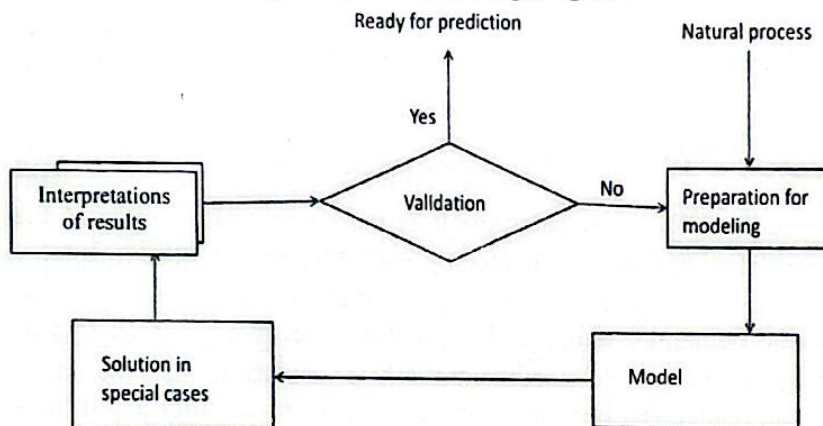
***** THE END *****



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- Use separate sheet for each section.

Section-AThere are **FOUR** questions in this section. Answer any **THREE** questions**Marks**

1. (a) Distinguish between modeling and mathematical modeling. Discuss briefly the process of mathematical modeling using the following diagram **03**



- (b) 'Conservation law plays a pivotal role in constructing a mathematical model'-Explain with your arguments. **03**
- (c) State Newton's cooling/heating law. Suppose that a chicken is removed from an oven, its temperature is then measured at 300°F. Three minutes later, if its temperature is 200°F, then how long will it take for the chicken to cool off to a room temperature of 70°F. **04**
2. (a) State the Malthusian growth law for a single species population. Construct an exponential growth model for a single species population. Discuss its solution and limitations, if any. **04**
- (b) Explain migration and harvesting in ecology. Suppose migration is considered and let e be the migration rate. Then for harvesting a renewable resource, an ecological model can be formulated as $\frac{dx}{dt} = rx\left(1 - \frac{x}{K}\right) - ex$. Find an analytical solution of the model and discuss the stability at equilibrium points. **06**
3. Consider a population of fish that is growing logistically and that is being harvested, assuming the catch of fish per unit effort is proportional to the stock level N . Discuss the model applying high-mortality harvesting and severe overfishing. Also, explain maximum sustainable yield at critical depensation. **10**
4. (a) Define linear and non-linear autonomous and non-autonomous dynamical systems. Explain the coupled and non-coupled differential equations. Discuss the equilibrium points and stability at equilibrium points. **04**
- (b) A two species competition model with linear autonomous systems can be formulated as $\frac{dx}{dt} = 3x - y$, $\frac{dy}{dt} = -2x + 2y$, where $x(0) = 90$ and $y(0) = 150$ and t is measured in years. (i) Determine the populations $(x(t), y(t))$ at all future times t . (ii) What will be the populations after six months? (iii) When will the species y be eliminated? **06**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

5. (a) Discuss the importance of delay in epidemic model. Incorporating delay in the systems, formulate a basic epidemic SIR model and solve the model with stability analysis. 04
 (b) A nonlinear prey-predator model with Holling type II response function takes the form 06

$$\dot{x}(t) = rx \left(1 - \frac{x}{K} \right) - \frac{sxy}{1 + sHx}, \quad \dot{y}(t) = \frac{esxy}{1 + sHx} - my.$$
 Let s be the search rate, P be the predation rate per predator, f be the fraction of time spent for searching and H be the time needed to handle one prey, then show that $P = \frac{sx}{1 + sHx}$. Also test the stability of the model at the equilibrium points.

6. (a) Solve the following delay differential equation on the interval $[0, 2]$ $\frac{dx}{dt} = x(t-1) + x(t)$, 04
 with initial condition $x(t) = 0$ for all $t \in [-1, 0]$.
 (b) Cooke proposed a delayed vector disease model given by $\frac{dy}{dt} = by(t-\tau)[1-y(t)] - ay(t)$, 06
 where $y(t)$ is the infected population, b is the contact rate and a is the cure. The discrete time delay τ gives the incubation period before the disease agent can infect a host. Cooke assumed that the total population is constant and scaled so that $x(t) + y(t) = 1$, $x(t)$ being the uninfected population. He also assumed that and infected population is not subject to death, immunity or isolation. (i) Find the steady state solution of the model. (ii) Find the condition for linear stability about the steady state solution. (iii) Obtain the numerical solution for (a) $a = 5.8, b = 4.8(a > b), \tau = 5$ and (b) $a = 38, b = 4.8(a < b), \tau = 5$. (iv) Compare and comment on the graphs.

7. (a) A rod of length 10 cm, whose sides are insulated, is kept at temperature 0°C and 100°C at its ends A and B respectively, until the steady state condition prevails. The temperature at end A is suddenly increases at 20°C and at end B, it is decreased 60°C . formulate a mathematical model of the given situation and obtain the temperature function at time t . 05
 (b) Derive the one-dimensional wave equation considering a homogeneous string of length L , tied at both ends such that the string offers no resistance due to bending. 05

8. Write down some basic features of zoonotic diseases. Is Nipah Virus (NiV) infection a zoonotic disease? Formulate an SEIR model of NiV infection using bilinear incidence rate in absence of permanent immunity and discuss the stability analysis in vein of basic reproduction number R_0 . 10

***** THE END *****

Khulna University, Khulna
Mathematics Discipline
4th Year, Term-I, Examination Session: 2023-2024
Course No: Math-4103
Full Title of Course: Mathematical Modeling
Full Marks: 60, Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

1. (a) Distinguish between modeling and mathematical modeling. Discuss briefly the processes of mathematical modeling using the flow diagram. 03
 (b) Discuss the 'conservation law' in constructing a mathematical model. 03
 (c) State Newton's cooling/heating law. Suppose that a chicken is removed from an oven, its temperature is then measured at 300°F. Three minutes later, if its temperature is 200°F, then how long will it take for the chicken to cool off to a room temperature of 70°F. 04

2. (a) State and explain Exponential growth (Malthus model) and Logistic growth (Verhulst Model) model. 05
 (b) Solve the logistic differential equation $\frac{dX}{dt} = rX\left(1 - \frac{X}{k}\right)$ given the initial condition $X(0) = x_0$. 05

3. (a) What is carrying capacity? Using this, derive a logistic growth model for single species population. Discuss the advantages of such model for long term prediction. Find an analytical solution of the model and discuss the stability at equilibrium points. 06
 (b) Describe the function of artificial kidney machine. Obtain the solution of a pharmacokinetic model for kidney dialysis. 04

4. (a) Describe linear and nonlinear autonomous and non-autonomous dynamical systems. Explain the coupled and non-coupled differential equations. Discuss the equilibrium points and different stability at equilibrium points. 04
 (b) A two species competition model with linear autonomous systems can be formulated as 06

$$\frac{dx}{dt} = 3x - y$$

$$\frac{dy}{dt} = -2x + 2y$$
 where $x(0) = 90$ and $y(0) = 150$ and t is measured in years.
 (i) Determine the populations $(x(t), y(t))$ at all future times t .
 (ii) What will be the populations after six months?
 (iii) When will the species y be eliminated?

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

5. (a) Discuss the importance of delay in epidemic model. Incorporating delay in the systems, formulate a basic epidemic SIR model and solve the model with stability analysis. 04
- (b) Define epidemic, endemic and pandemic infectious diseases with examples. What is basic reproduction number? Discuss some basic assumptions of epidemic models. What do mean by the horizontal transmission and vertical transmission in case of disease propagation? 06
6. (a) What is deterministic compartmental (SIR) models? Obtain the numerical solution of the following SIR model: $\frac{dS}{dt} = -\beta \frac{SI}{N}$, $\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I$, $\frac{dR}{dt} = \gamma I$, $S(0) = S_0$, $I(0) = I_0$, $R(0) = R_0$. Here, β is the rate of transmission of infection and γ is the rate of recovery. 05
- (b) What are the basic assumptions of a two-species prey-predator interacting model? Find the exact solution of two-species Lotka-Volterra prey-predator model. 05
7. (a) Explain the Delay Differential Equations with its kinds and state the important applications of delayed differentiations for different types of delays. What are the important differences between ordinary differential equations and delayed differential equations? 05
- (b) Formulate an SIR epidemic model and discuss the stability of the disease in terms of basic reproduction number R_0 . 05
8. Suppose a company makes two products (say P and Q) using two machines (say A and B). Each unit of P that is produced requires 50 minutes processing time on machine A and 30 minutes processing time on machine B. Each unit of Q that is produced requires 24 minutes processing time on machine A and 33 minutes processing time on machine B. Machine A is going to be available for 40 hours and machine B is going to be available for 35 hours. The profit per unit of P is \$25 and the profit per unit of Q is \$30. Company policy is to determine the production quantity of each product in such a way as to maximize the net profit given that the available resources should not exceeded. Formulate this problem into a mathematical model with linear programming and solve the model to satisfy the company's policy. 10

***** THE END *****

Khulna University
 Science, Engineering and Technology School
 Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**
 Year: **Fourth**, Term: **I** Examination-2026; Session: **2024-2025**
 Course Code: **0541 12 Math 4103**
 Course Title: **Mathematical Modeling**
 Total Marks: **60**, Time: **03** Hours, Credit: **03**



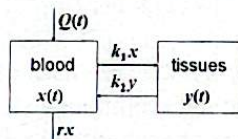
- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

Marks

1. (a) Explain modeling and mathematical modeling in terms of dynamical systems. Discuss the different processes of mathematical modeling with appropriate flow chart. [L1; CLO1] **03**
- (b) How balance principle is preserved in mathematical modeling? Explain with your argument. [L2; CLO1] **04**
- (c) Incorporating all possible limitations of unbounded growth model, derive a logistic growth model for single species population. Solve the model with stability analysis. [L3; CLO2] **03**
2. (a) Suppose the population $p(t)$ is given by the exponential growth law. Growth rate r is positive. Let T be the amount of time required for the starting population p_0 to double. Then show that $T = \frac{\log_e 2}{r}$ which is independent of p_0 . [L2; CLO2] **06**
- (b) How mathematical model can be applied to kidney dialysis process? Describe and formulate an artificial kidney dialysis model and discuss the solution of the model. [L3; CLO2] **04**
3. (a) Explain pharmacokinetics. Describe and formulate a pharmacokinetic model using the following diagram. Solve the model with all possible equilibrium points and stability analysis. [L2; CLO3] **06**



- (b) A symbiotic linear model can be formulated by the following governing equations. **04**

$$\frac{dx}{dt} = -x + 2y, \frac{dy}{dt} = 2x - y$$
 where $x(0) = 100$ and $y(0) = 300$. Check the nature of stability at $(0,0)$. Determine the populations $(x(t), y(t))$ at all future times t . What will be the populations after six months? When will the species y be eliminated? [L3; CLO3]
4. (a) State and explain two species Lotka-Volterra competition model. [L1; CLO4] **05**
- (b) What is traffic flow? What are the objectives of traffic flow modeling. Numerically solve the LWR (Lighthill, Whitham and Richards) traffic flow model $\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v)}{\partial x} = 0$, $v = v(\rho)$. Given, Initial conditions: $\rho(x, 0) = \rho_0(x)$ and Boundary conditions: $\rho(0, t) = \alpha$, $\rho(L, t) = \beta$. [L4; CLO4] **05**

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | Marks |
|--------|---|-------|
| 5. (a) | Discuss the importance of discrete model. Construct an SIR type continuous model of disease dynamics. Convert this continuous model in the form of difference equation and solve with stability analysis. [L2; CLO5] | 06 |
| (b) | Suppose "bugbox" is a simple computer simulation for structured population and it functions as a biology lab for a virtual world . For this "bugbox" model, let L_t be the number of larvae at time t , J_t be the number of juveniles at time t and A_t be the number of adults at time t . Formulate a structured population model and discuss the solution of the model with stability analysis. [L4; CLO5] | 04 |
| 6. | Consider the pre-predator model with functional response $\frac{dx}{dt} = rx \left(1 - \frac{x}{k}\right) - bxy$, $\frac{dy}{dt} = cxy - dy$, where x is prey and y is predator. Examine the positivity and analyze its stability. [L3; CLO6] | 10 |
| 7. | Discuss different <i>force of infections</i> in case of disease transmission. An SIRS epidemic model with nonlinear saturated incident rate can be constructed as $\frac{dS}{dt} = (1-p)A - \frac{\beta SI}{1+\alpha I} - dS + \gamma R$, $\frac{dI}{dt} = pA + \frac{\beta SI}{1+\alpha I} - (d+m)I$, $\frac{dR}{dt} = mI - (d+\gamma)R$, where $(1-p)A$ and pA are the constant recruitments of S and I individuals. Describe the model with a schematic diagram. Find the equilibrium points and test the stability at disease free equilibrium point. Explain basic reproduction number. [L5; CLO5] | 10 |
| 8. | Consider two different types of foodstuffs, say, F_1 and F_2 . Assume that these foodstuffs contain vitamins V_1 , V_2 and V_3 respectively. Minimum daily requirements of these vitamins are 1mg of V_1 , 50mg of V_2 and 10mg of V_3 . Suppose that the foodstuffs F_1 contains 1mg of V_1 , 100mg of V_2 and 10mg of V_3 , whereas foodstuffs F_2 contains 1mg of V_1 , 10mg of V_2 and 100mg of V_3 . Cost of one unit of foodstuffs F_1 is \$1 and that of F_2 is \$1.5. Formulate a linear programming model to find the minimum cost diet that would supply the body at least minimum requirements of each vitamin. [L3; CLO6] | 10 |

***** THE END *****

Khulna University
 Science, Engineering and Technology School
 Discipline: **Mathematics**, Program: **B.Sc. (Hons.)**
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 Total Marks: **60**, Time: **03** Hours, Credit: **03**



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- Use separate answer script for each section.

Section-A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | Marks |
|----|---|--------------|
| 1. | (a) Discuss the different processes of mathematical modeling with appropriate flow chart. | 03 |
| | (b) How is balance principle preserved in mathematical modeling? Explain with your argument. | 03 |
| | (c) Discuss the possible limitations of unbounded growth model. Derive a logistic growth model for single species population. Solve the model with stability analysis. | 04 |
| 2. | (a) State the Malthusian growth law for a single species population. Construct an exponential growth model for a single species population. Discuss its solution and limitations, if any. | 04 |
| | (b) Explain migration and harvesting in ecology. Suppose migration is considered and let e be the migration rate. Then for harvesting a renewable resource, an ecological model can be formulated as
$\frac{dx}{dt} = rx \left(1 - \frac{x}{K}\right) - ex.$ Find an analytical solution of the model and discuss the stability at equilibrium points. | 06 |
| 3. | (a) Construct a fishery management model for single species population with harvesting effort.
Solve the model and discuss the stability of the solution. | 05 |
| | (b) Identify and solve the linear population model
$x' = x + y$ $y' = -x + y$ where $x(0) = y(0) = 1000$.
Determine all future populations; when will the prey population be eliminated? | 05 |
| 4. | A predator-prey model in a two species ecosystem with <i>Holling type- II</i> can be formulated as
$x'(t) = rx \left(1 - \frac{x}{K}\right) - \frac{sxy}{1 + sHx}$ $y'(t) = \frac{esxy}{1 + sHx} - my$ where, x is the density of prey and y is the predator density. Verify the model for well posedness and discuss the stability at equilibrium points and sketch the phase portraits. Describe the trajectories in xy plane. | 10 |

Section-B

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | |
|----|--|-----------|
| 5. | (a) Discuss the importance of discrete model. Construct a SIR type continuous model of disease dynamics. Convert this continuous model in the form of difference equation and solve with stability analysis. | 05 |
| | (b) Let x and y denote the two conflicting nations, and let $x(t)$ and $y(t)$ be the war potentials of nation x and y respectively. So, the rate of change of war potential can be given by | 05 |

$$\frac{dx}{dt} = ky - \alpha x + g$$

$$\frac{dy}{dt} = lx - \beta y + h$$

where $\alpha, \beta, k, l, g, h$ are positive constant. Here α, β depends on the resource constraints of the respective nations; k, l depends on the war readiness of opponent nations and g, h are depend on the dissatisfaction of one nation against of the order. Discuss the solution of the model.

6. (a) Discuss the importance of delay in epidemic model. Incorporating delay in the systems, formulate a basic epidemic SIR model and solve the model with stability analysis. 04
 (b) A nonlinear prey-predator model with Holling type II response function takes the form 06

$$\dot{x}(t) = rx \left(1 - \frac{x}{K}\right) - \frac{sxy}{1 + sHx}$$

$$\dot{y}(t) = \frac{esxy}{1 + sHx} - my$$

Let s be the search rate, P be the predation rate per predator, f be the fraction of time spent for searching and H be the time needed to handle one prey, then show that $P = \frac{sx}{1+sHx}$. Also test the stability of the model at the equilibrium points.

7. Discuss different *force of infections* in case of disease transmission. An SIRS epidemic model with nonlinear saturated incident rate can be constructed as 10

$$\frac{dS}{dt} = (1-p)A - \frac{\beta SI}{1+\alpha I} - dS + \gamma R, \frac{dI}{dt} = pA + \frac{\beta SI}{1+\alpha I} - (d+m)I, \frac{dR}{dt} = mI - (d+\gamma)R,$$
 where $(1-p)A$ and pA are the constant recruitments of S and I individuals. Describe the model with a schematic diagram. Find the equilibrium points and test the stability at disease free equilibrium point. Explain basic reproduction number.
8. Consider two different types of foodstuffs, say, F_1 and F_2 . Assume that these foodstuffs contain vitamins V_1, V_2 and V_3 respectively. Minimum daily requirements of these vitamins are 1mg of V_1 , 50mg of V_2 and 10mg of V_3 . Suppose that the foodstuffs F_1 contains 1mg of V_1 , 100mg of V_2 and 10mg of V_3 , whereas foodstuffs F_2 contains 1mg of V_1 , 10mg of V_2 and 100mg of V_3 . Cost of one unit of foodstuffs F_1 is \$1 and that of F_2 is \$1.5. Formulate a linear programming model to find the minimum cost diet that would supply the body at least minimum requirements of each vitamin. 10

***** THE END *****